HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and developed integration solutions between **MES platforms** and **enterprise applications**, coordinating with cross-functional teams including Automation, IT/OT, and Quality.
- Managed equipment connectivity and developed **APIs** and **middleware solutions** to facilitate manufacturing data flows within a regulated GMP environment.
- Supported MES system implementation, configuration, and testing activities, ensuring readiness for operational deployment.
- Troubleshot and resolved complex integration issues involving **MES**, **databases**, and **automation systems**, minimizing production downtime.
- Developed comprehensive technical documentation, including **functional specifications**, **interface designs**, and **test scripts** for validation purposes.
- Contributed to the definition and implementation of integration standards and best practices for manufacturing systems.
- Containerized integration services using **Docker** and deployed them to **AWS EC2** instances, ensuring reliable operation within the cloud environment.

SKILLS

MES Platforms – PAS-X, Syncade, PharmaSuite

Integration Technologies – APIs, Middleware, OPC/OPC UA

Manufacturing Systems – PLC, SCADA, Historian, IT/OT

Databases – SQL, PostgreSQL

Compliance & Validation – GMP, GxP, Data Integrity, CSV, Commissioning, Qualification

Cloud & Containerization – AWS, EC2, Docker

Development Tools – Java Spring Boot, Python, Jenkins, Git

PROJECTS

MES Integration for Pharmaceutical Manufacturing

- Developed custom Java Spring Boot services to act as middleware, integrating a **PharmaSuite MES** with an existing **Oracle database** containing batch production data.
- Designed and implemented RESTful **APIs** for real-time data exchange between the MES and an upstream enterprise resource planning system.
- Authored detailed **functional specification documents** and executed unit tests for all developed integration components.

Automation Data Flow Enhancement

- Engineered Python scripts to parse and transform data from **OPC UA** sources, preparing it for ingestion into a central **SQL database** for operational analytics.
- Automated report generation using Python scripts that queried the database and produced daily summaries of equipment status and production throughput, reducing manual reporting time by **40%**.
- Collaborated with the automation team to define data schemas and ensure seamless data flow from PLCs to the MES.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)